



CONTACT

EMAIL

hsandmann@ieee.org

WEB

hsandmann.github.io

GITHUB

hsandmann

LINKEDIN

in/hsandmann

PROJECT

graphole.com

SKILLS

Languages

Java Python Go Kotlin C

C++ JavaScript SQL

Machine Learning & AI

Deep Learning Neural Networks

Computer Vision NLP PCA SVM

Clustering

Backend & Architecture

Spring Boot Spring Cloud

Microservices Design Patterns

Software Architecture

Data & Infrastructure

PostgreSQL Redis Kafka

RabbitMQ Docker Kubernetes

Spark MongoDB Neo4j

Domains

Banking & Fintech

Cryptography (PKI/HSM)

High-Performance Computing

POS/ERP

LANGUAGES

Portuguese Native

English Advanced

German B1

Humberto Sandmann

Computer Scientist & Professor

PROFILE

Senior software architect and artificial intelligence engineer with 20+ years building high-stakes systems across banking, fintech, and retail — core banking, cryptography (PKI/HSM), high-performance computing, computer vision, and cloud-native microservices. D.Sc. in neural computing; also a university professor teaching software design, AI, and distributed systems.

EXPERIENCE

Founder · H-Sandmann

2006–present

Independent consultancy in machine learning and software architecture.

Machine Learning

Software Architecture

Co-Founder · SelsanTech

2012–2023

Co-founded a venture delivering machine-learning and software-architecture solutions.

Machine Learning

Software Architecture

Senior Software Architect & Machine Learning Engineer · RockSpoon

2018–2019

Architected POS/ERP/payroll platforms and computer-vision/ML features for a restaurant-tech company.

POS

ERP

JavaEE

PostgreSQL

OpenCV

Computer Vision

Go

Android

Kotlin

Senior Consultant, Computational Optimization · Banco Votorantim

2015–2016

Return-on-equity optimization on high-performance computing infrastructure for a major bank.

RoE

High-Performance Computing

J2EE

Spring Boot

Jenkins

Test Automation

Senior Consultant, Computational Optimization · TecBan

2011–2011

ATM network optimization with in-memory databases and high-performance computing.

ATM

J2EE

In-Memory DB

High-Performance Computing

Senior Software Engineer / Architect · ABN Amro Bank

2006–2007

Core-banking, stock-market and trading systems with cryptography and PKI.

Core Banking

Trading

Cryptography

HSM

BouncyCastle

PKI

PKCS

Java

Servlet

Senior System Analyst · Politec

2005–2006

Senior Software Engineer / Architect · EverSystems

2004–2005

Senior System Analyst · Central Online

2002–2003

Senior System Analyst · Datamace

1999–2002

Computer Programmer · Heraeus Electro-Nite

1998–1999

EDUCATION

D.Sc. · Electrical Engineering — Polytechnic School, University of São Paulo 2007–2012

M.Sc. · Electrical Engineering — Polytechnic School, University of São Paulo 2003–2006

B.Sc. · Computer Science — Faculdade de Engenharia Industrial (FEI) 1999–2002

Full academic CV, publications and teaching record at hsandmann.github.io